Predicting Federal Reserve Rate Decisions from FOMC minutes using FinBERT

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1 Introduction

1.1 Financial Motivation

Interest rates are among the most consequential variables in finance, influencing everything from consumer credit to corporate borrowing cost. The Federal Reserve (Fed) is the world's most closely watched central bank. It shapes monetary conditions not only through rate decisions themselves but through deliberate use of language to signal future policy intentions - a practice known as forward guidance [3]. Lucca & Trebbi (2009) [6] demonstrated that the content of FOMC statements contains significant information about future policy rate decisions, leading actual rate changes by more than a year - indicating that the minutes contain potential signal. Released three weeks after each meeting, the minutes provide a rich source of forward-looking information that precedes the subsequent rate decision. This paper focuses on a three-head classification task, predicting whether the Fed's next decision will be a hike, hold or cut. We apply FinBERT [1], a transformer-based language model pre-trained on financial text, to these minutes when performing the classification.

1.2 Why Machine Learning

Human analysis of FOMC minutes is subject to inconsistent interpretation and is impractical at larger scale when attempting to identify patterns across decades of data. Early Natural Language Processing (NLP) approaches applied Bag-of-Words (BoW) methods to financial text, demonstrating that word frequencies carry meaningful signal [2], but also revealing that such methods misclassify common financial terminology and ignore word order entirely [5]. These limitations are particularly significant for central bank language, where context of phrases carry fundamentally different policy implications that are not captured by word counts. Later developments introduced the concept of attention mechanisms and transformers[7] which directly addresses these flaws by developing situational dependencies between words in a sequence simultaneously. Bert [4] implements this into a general purpose pre-trained model. However, its training on general corpora showed limited performance on the specialised vocabulary such as that of financial institutions. FinBERT resolved this by pre-training on approximately 29 million words of financial text drawn from Reuters and Bloomberg, enabling accurate representations of the technical language used in finance, making it a natural methodological choice in this task.

2 Methodology

2.1 FOMC and Interest Rates

The Federal Market Open Committee (FOMC) meet approximately eight times per year to assess monetary policy circumstances. The minutes of each meeting are published three weeks after the meeting date. To avoid lookahead bias, this study uses minutes from meeting T to predict the rate decision at meeting T+1. Data from 1993 is used, as earlier minutes adopt a non-standardised format. This is also the era that central banks introduced forward guidance[3], where meetings involve speculation regarding future movements in interest rates alongside reasoning for current decisions.

2.2 Fine-Tuned FinBERT

2.2.1 FinBERT

FinBERT is a domain-adapted extension of BERT, a transformer architecture built entirely from stacked encoder layers, enabling bidirectional processing of input text. BERT employs bidirectional self-attention, allowing each token to attend to all other tokens in the sequence simultaneously. Each token is informed by its full surrounding context, helping to produce contextual representations - which is essential for central bank communication.

$$Attention(Q,K,V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

where $Q, K \in R^{n\times d_k}$ and $V \in R^{n\times d_v}$ are the query, key and value matrices, n is the sequence length and d_k is the key dimensionality. The scaling factor $\sqrt{d_k}$ prevents large dot products from pushing the softmax function into regions with vanishing gradients

The base model was pre-trained by Google on BookCorpus and Wikipedia through masked language modelling, whereby 15% of input tokens are randomly masked and the model is trained to reconstruct them. FinBERT extended this through continued training on financial corpora. The architecture consists of 12 Transformer encoder layers, each with 12 multi-headed attention heads, a hidden dimension of 768 and a feedforward dimension of 3,072, totalling roughly 110 million parameters. For sequence classification, a dense layer is added to the `[CLS]` token representation produced by the final encoder.

$$\hat{y} = softmax(W \cdot h_{[CLS]} + b)$$

where $h_{[CLS]} \in R^{768}$ is the hidden state of the `[CLS]` token from the final encoder layer, W is the learned weight matrix of the classification head and b is the bias vector. The output $\hat{y}$ is a probability distribution over the three classes {`hike, hold, cut`}.

2.2.2 Model Adapted FinBERT

To adapt FinBERT for rate decision classification, several changes were performed. The original three-class sentiment head (*positive, neutral, negative*) was replaced with a three-class policy head (*hike, hold, cut*) such that the pre-trained encoder weights are inherited while the output layer is trained from scratch. Inputs were tokenized with `BertTokenizer` with a maximum sequence length of 128 tokens, with padding and truncation applied where necessary.

Hyperparameters were selected via grid search, evaluating all combinations of learning rates $\{1\times10^{-5}, 2\times10^{-5}, 3\times10^{-5}\}$ and batch sizes $\{16, 32\}$ with the best configuration chosen by the maximum validation F1-score. The tests were fixed over three epochs, chosen following research from Devlin et al. (2018) who suggest that over 4 epochs may result in overfitting. This is also consistent with computational and time constraints. Weighted cross-entropy loss was applied to address class imbalance, assigning stronger penalties to classes with higher frequency, the intention being to discourage the model from defaulting to the majority *hold* class.

$$\mathscr{L} = -\sum_{c=1}^{C} w_c \cdot y_c \log(\hat{y}_c)$$

where $w_c = \frac{N}{C \cdot N_c}$ is the weight assigned to class c, N is the total number of training samples, N_c is the number of samples in class c, and $C = 3$ is the number of policy classes. Gradient clipping with a maximum norm of 1.0 was applied to stabilise training, protecting against the large gradient updates that can arise when fine-tuning a large pre-trained model on a relatively small dataset. The optimal configuration of a learning rate of 1×10^{-5} and a batch size of 16 was selected due to it having the highest validation macro F1 and was used for all subsequent evaluation.

As full meeting minutes exceed the 512-token limit of the tokeniser, documents were segmented into individual sentences prior to inference. Sentence-level predictions were then aggregated to meeting level via majority vote, since the target label is a meeting-level outcome.

2.3 Baseline Model: TF-IDF with Logistic Regression

TF-IDF was used in combination with logistic regression as a baseline. It follows a bag-of-words approach, identifying terms that appear regularly across documents whilst nullifying common everyday terminology. It is composed of two components: Term Frequency (TF) which measures how often a term appears in a document d:

$$TF(t,d) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}} \tag{1}$$

Inverse Document Frequency (IDF) which down weights terms that appear in many documents:

$$IDF(t,D) = \log \frac{|D|}{|\{d \in D : t \in d\}|} \tag{2}$$

The TF-IDF score is the product of these two components. Logistic regression then takes these vectors as input and learns a linear boundary for classification. Logistic regression then maps these vectors to class probabilities:

$$P(y = c \mid \mathbf{x}) = softmax(\mathbf{W}\mathbf{x} + \mathbf{b})$$

where $\mathbf{x}$ is the TF-IDF vector representation of the meeting minutes and $\mathbf{W}$, $\mathbf{b}$ are learned parameters.TF-IDF deliberately ignores word order and context, any performance gap between this baseline and FinBERT can be specifically attributed to contextual understanding - allowing for clean interpretation of what the transformer architecture contributes to this task.

2.4 Data

2.4.1 Data Sources and Ingestion

The data set consists of two components. The primary source was FOMC meeting minutes, scraped directly from the Federal Reserve website, covering February 1993 to January 2026, resulting in around 277 meetings at approximately eight per year. The website lists minutes, transcripts and press conferences. Ultimately, minutes were selected due to their consistent structure, direct provision of information, and non-inclusion of conversational noise present in transcripts such as procedural exchanges. The date of 1993 was deliberately chosen. Alongside the reasons mentioned earlier, the language predates data that FinBERT was trained on, risking degraded performance.

Rate decision labels were obtained via the Federal Reserve Economic Data (FRED) API, retrieving the federal funds rate over the same period. Each rate observation was matched to its corresponding meeting on a datetime index, then categorised as hike, hold or cut by comparing each value to its predecessor.

2.4.2 Preprocessing

Each set of minutes was pre-processed to remove attendance lists and administrative boilerplate, retaining only relevant policy discussions. As full meeting minutes exceed FinBERT's 512-token limit, each meeting was split into individual sentences prior to classification, with predictions subsequently aggregated via majority vote at the meeting level. After preprocessing, 54,876 sentences remained from 277 meetings. Training was conducted on a T4 GPU via Google Colab.

2.4.3 Train, Validation and Test Split

Chronological splitting was applied as opposed to random shuffling - to prevent data leakage in the training set and retain temporal validity. The data was split chronologically into training, validation and test sets which corresponds to the periods 1993-2017, 2018-2021, and 2022-2026 respectively.

Split	Period	Hold	Hike	Cut	Total
Train	1993–2017	21,953	9,982	6,522	38,457
Validation	2018–2021	5,575	1,706	1,458	8,739
Test	2022–2026	2,947	2,907	1,740	7,594

Table 1: Sentence-level class distribution across chronological splits. Hold meetings constitute the majority of observations across all splits, with cut decisions underrepresented, motivating the use of weighted loss during training.

3 Results

3.1 Exploratory Analysis

Prior to fine-tuning, pre-trained FinBERT sentiment scores were aggregated to meeting level to assess whether linguistic signal exists in FOMC minutes. The mean policy score exhibits a monotonic relationship with subsequent rate decisions (Table 2), confirming that Fed language is measurably more optimistic in meetings preceding hikes than cuts.

This monotonic ordering motivates the use of fine-tuned FinBERT for direct policy classification in section 3.2.

Next Decision	Mean Policy Score
Hike	0.429
Hold	0.394
Cut	0.363

Table 2: Mean meeting-level policy score by subsequent rate decision.

3.2 Training Performance

Figures 1 and 2 present the training loss and validation macro F1 across epochs for the optimal hyperparameter configuration (learning rate 1×10^{-5}, batch size 16). Training loss decreases consistently from 1.00 at epoch 1 to 0.70 at epoch 3, reflecting stable learning. Validation macro F1 peaks at 0.42 at epoch 2 before marginally declining.

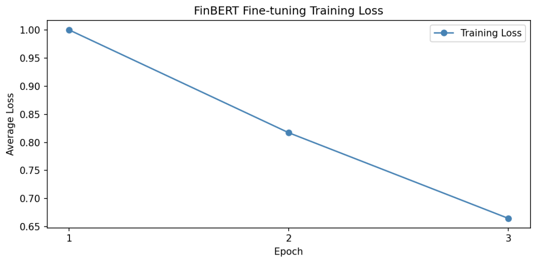

Figure 1: Training loss by epoch

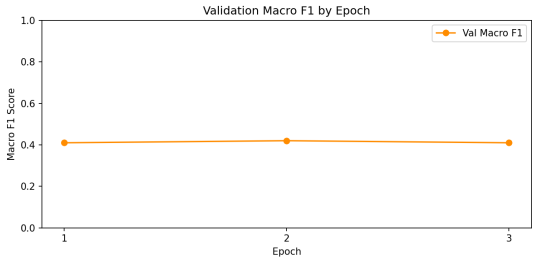

Figure 2: Validation macro F1 by epoch

3.3 Test Performance

Table 3 presents meeting-level performance on the held-out test set (2022–2026) for both models alongside the random prediction baseline.

Model	Accuracy	Macro F1	Hold F1	Hike F1	Cut F1
FinBERT (fine-tuned)	0.41	0.19	0.58	0.00	0.00
TF-IDF + Logistic Regression	0.45	0.26	0.00	0.17	0.60
Random baseline	—	0.33	—	—	—

Table 3: Meeting-level classification results on the 2022–2026 test set (32 meetings: 13 hold, 11 hike, 8 cut). Macro F1 weights each class equally regardless of frequency. Both models perform below the random baseline of 0.33.

The fine-tuned FinBERT model collapses entirely to predicting *hold* for all 32 test meetings, correctly identifying the 13 hold meetings (F1 = 0.58) while failing on every hike and cut. The TF-IDF baseline exhibits a different failure, it identifies no hold meetings at all, correctly predicts 1 of 11 hikes (F1 = 0.17), and classifies all cuts correctly (F1 = 0.60). Both models fall below the random baseline macro F1 of 0.33, indicating that neither generalises to the test regime. Confusion matrices for both models are presented in Figures 3 and 4.

At the sentence level, the fine-tuned model retains some discriminative capacity during validation, achieving a macro F1 of 0.41 across all three classes, meaningfully above the random baseline. However, this signal does not survive majority-vote aggregation to the meeting level on the test period.

4 Discussion

4.1 Interpretation of Results

The results reveal a fundamental failure of the fine-tuned FinBERT model to generalise to the test period. Despite demonstrating genuine learning during validation, all meeting level predictions collapsed to *hold*, achieving a macro F1 of 0.19 compared to the TF-IDF baseline's score of 0.26.

The per-meeting breakdown highlights the nature of this failure. The model correctly identifies all 13 hold meetings in the test set but misclassifies every hike and cut. Examining at the sentence-level, predictions provide further insight. Of 7,594 test sentences,

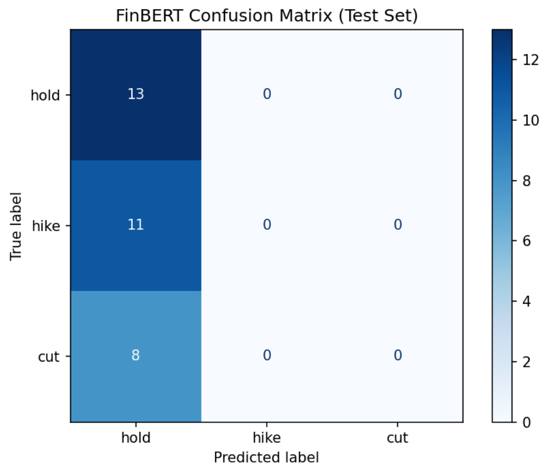

Figure 3: FinBERT confusion matrix

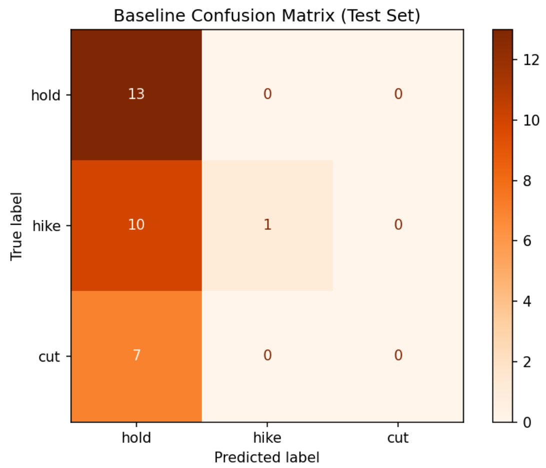

Figure 4: Baseline confusion matrix

5,422 were classified as hold, 1,785 as hike and only 387 as cut. Despite this distribution the model showed somewhat strong classification capacity at sentence level, however the aggregation caused an overwhelming majority at meeting level.

The validation results tell a different story. Across three epochs the model demonstrated consistent learning with macro F1 improving from 0.389 to 0.405 and sentence-level performance reasonably distributed across all three classes. The divergence between the test and validation sets strongly points to the distribution shift rather than to a learning failure. The test period presents a distinctly different regime - this is due to the test period being the most intense tightening cycle in decades, which the model had little exposure to in training.

This impact is magnified by the class imbalance in training data. Hold meetings constitute the majority of observations. Despite class weighting the model appears to have caused a default to the path of least resistance under an unfamiliar regime.

Sentence-level validation achieves a macro F1 of 0.41, meaningfully above the random baseline of 0.33 which demonstrates that FinBERT extracts genuine policy signal from the FOMC text. However, at the meeting-level, the aggregation on the 2022-2026 test period reveals a temporal generalisation challenge - both the fine-tuned model and the TF-IDF baseline default to hold prediction.

4.2 Limitations

There are several limitations of this methodology. A fundamental issue is the non-stationarity of macroeconomic regimes, models trained on one economic state may not generalise well to structurally different periods.

Secondly, the sentence-level classification approach discards document-level structure and ordering. Certain sections of documents will be more policy relevant than others, this is not picked up during majority vote aggregation which gives each sentence an equal importance. This means core policy content will not be reflected properly in regards to distribution.

Although class weighting was applied to address imbalance on the training set, the hold class remains inherently difficult to classify. The use of neutral, impartial language lacks distinctive patterns that can be found to characterise hike or cut language making it the hardest text to detect alone.

Lastly, earlier minutes from our dataset use less standardised language than modern day and predate the Fed's formal adoption of forward guidance as a policy tool, meaning FinBERT may be less specialised during this period of time.

4.3 Future Work

Several applications could be used to address some of the limitations in this work. The first being a hierarchical attention approach, classifying at sentence level before aggregating through a document level attention mechanism - this would allow the model to learn which sentences carry stronger signal relative to policy decision.

During validation a walk-forward approach could be used where the model is retrained on a rolling window which would allow for a more rigorous assessment of out-of-sample performance and permit a more robust performance against varying economic regimes.

Although minutes and FOMC statements can be argued to carry a valuable signal, it would be worth pairing this with economic indicators, which can provide a quantitative perspective that minutes do not strongly provide.

Finally, extending this framework to that of other central banks including the European central bank or Bank of England would test the commonness of linguistic patterns across a variety of institutions.

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